⚡ LeadPulse

Production Deployment Guide

Step-by-step instructions to go live

FastAPI · Supabase · Meta Graph API · Railway · Vercel

Version 1.0 · 2024

Overview

This guide walks you through deploying LeadPulse from your downloaded zip file to a fully live, production URL. You will complete five areas:

#	Area	Time	Service Used
1	Supabase — Database & Auth	10 min	supabase.com (free)
2	Meta App — Facebook Webhooks	15 min	developers.facebook.com
3	Backend — FastAPI API	10 min	Railway.app (~$5/mo)
4	Frontend — React App	5 min	Vercel.com (free)
5	Connect & Test Everything	10 min	—

ℹ️ Total estimated time: 50 minutes for a first deployment. Subsequent deploys take under 5 minutes using the Makefile.

Prerequisites

Before you start, make sure you have the following installed and available:

- **nodejs.org/en/download**Node.js 20+ and npm —
- **python.org/downloads**Python 3.12+ —
- **git-scm.com**Git —
- **github.com**A GitHub account (free) —
- **supabase.com**A Supabase account (free) —
- **developers.facebook.com**A Meta Developer account (free) —
- **railway.app**A Railway account (free trial, then ~$5/mo) —
- **vercel.com**A Vercel account (free) —

Step 1 — Unzip & Push to GitHub

All deployment services (Railway, Vercel) pull code from GitHub. Push your project there first.

1.1 Unzip the project

```
# Unzip the downloaded file
$ unzip leadpulse-fullstack-final.zip
$ cd leadpulse-fullstack

# Verify the structure
$ ls
  backend/   frontend/   docker-compose.yml   nginx.conf   README.md   Makefile
```

1.2 Create a GitHub repository

1. Go to github.com and click the + button in the top-right corner.
2. Click New repository.
3. Name it leadpulse (or any name you prefer).
4. Set it to Private (your .env files are gitignored, but best practice is private).
5. Do NOT initialise with a README — the project already has one.
6. Click Create repository.

1.3 Push the code

```
$ cd leadpulse-fullstack
$ git init
$ git add .
$ git commit -m "Initial commit — LeadPulse fullstack"
$ git branch -M main
$ git remote add origin https://github.com/YOUR_USERNAME/leadpulse.git
$ git push -u origin main
```

✅ Your code is now on GitHub. Every deployment service will pull from this repo.

Step 2 — Set Up Supabase (Database)

Supabase is a managed PostgreSQL database with a built-in API and auth system. The free tier is more than enough for getting started.

2.1 Create a Supabase project

7. Go to supabase.com and sign in with GitHub.
8. Click New Project.
9. Choose your organisation (or create one).
10. Fill in the project name: leadpulse
11. Set a strong database password — SAVE THIS PASSWORD, you will need it.
12. Select the region closest to your users (e.g. ap-south-1 for India).
13. Click Create new project and wait ~2 minutes for it to provision.

2.2 Run the database schema

14. In your Supabase project, click SQL Editor in the left sidebar.
15. Click New query.
16. Open the file backend/supabase_schema.sql from your project folder.
17. Copy the entire contents and paste it into the SQL Editor.
18. Click Run (green button, top right). You should see Success. No rows returned.

> ✅ This creates all tables (workspaces, users, leads, notes, connected_pages), enums, indexes, RLS policies, the updated_at trigger, and the leads_daily_trend analytics RPC function.

2.3 Get your API keys

19. Click Settings (gear icon) in the left sidebar.
20. Click API under Configuration.
21. Copy and save the following three values:

Key Name	Where to find it	Used in
Project URL	Settings → API → Project URL	SUPABASE_URL
anon public key	Settings → API → Project API Keys → anon	SUPABASE_ANON_KEY
service_role key	Settings → API → Project API Keys → service_role (click reveal)	SUPABASE_SERVICE_ROLE_KEY

🚫 Never expose the service_role key on the frontend or in a public repository. It bypasses Row Level Security and has full database access.

Step 3 — Create a Meta Developer App

The Meta (Facebook) app is what allows LeadPulse to receive lead data via webhooks and let users connect their Facebook Pages.

3.1 Create the app

22. Go to developers.facebook.com and log in.
23. Click My Apps → Create App.
24. Select Other as the use case, then click Next.
25. Select Business as the app type, then click Next.
26. Fill in: App name: LeadPulse, Contact email: your email. Then click Create app.

3.2 Add required products

After the app is created, you land on the App Dashboard. Add two products:

27. Find Facebook Login → click Set up. Choose Web. Click Save.
28. Find Leads Access → click Set up. (If not visible, click + Add Product and search for it.)

3.3 Get your App ID and App Secret

29. Click Settings → Basic in the left sidebar.
30. Copy App ID — save as FACEBOOK_APP_ID.
31. Click Show next to App Secret — copy it and save as FACEBOOK_APP_SECRET.

3.4 Configure Facebook Login

32. In the left sidebar, click Facebook Login → Settings.
33. Under Valid OAuth Redirect URIs, add your frontend URL + /settings, for example: https://leadpulse.vercel.app/settings
34. Note: You will come back to this after deploying the frontend (Step 5). For now leave it and continue.
35. Click Save Changes.

3.5 Configure the Webhook

The webhook receives real-time lead events from Meta. You need your backend to be live first (Step 4), then return here.

> ℹ️ Come back to this section after completing Step 4 (Deploy Backend). You will need your live Railway backend URL.

36. In the left sidebar, click Webhooks.
37. Click Add Callback URL.
38. Callback URL: https://YOUR-RAILWAY-URL.railway.app/webhook/facebook
39. Verify Token: Enter the same random string you set in FACEBOOK_WEBHOOK_VERIFY_TOKEN in your .env (e.g. leadpulse_secret_2024).
40. Click Verify and Save. Meta will call your webhook endpoint to verify it — this must succeed.
41. After verifying, find the leadgen field and click Subscribe.

Step 4 — Deploy the Backend (Railway)

Railway is the simplest way to host a Python/FastAPI backend. It reads your Dockerfile and deploys automatically on every git push.

4.1 Create a Railway project

42. Go to railway.app and sign in with GitHub.
43. Click New Project → Deploy from GitHub repo.
44. Select your leadpulse repository.
45. When asked for a root directory, enter: backend
46. Railway will detect the Dockerfile automatically. Click Deploy.

4.2 Set environment variables

This is the most important step. Railway needs all your secrets to run the API.

47. In your Railway project, click on the deployed service.
48. Click Variables in the top tab bar.
49. Click New Variable and add each of the following:

Variable Name	Value
SUPABASE_URL	https://xxxx.supabase.co (from Step 2.3)
SUPABASE_ANON_KEY	your anon key (from Step 2.3)
SUPABASE_SERVICE_ROLE_KEY	your service_role key (from Step 2.3)
SECRET_KEY	run: openssl rand -hex 32 and paste the output
FACEBOOK_APP_ID	your App ID (from Step 3.3)
FACEBOOK_APP_SECRET	your App Secret (from Step 3.3)
FACEBOOK_WEBHOOK_VERIFY_TOKEN	a random secret string you choose, e.g. leadpulse_secret_2024
ENVIRONMENT	production
DEBUG	false
ALLOWED_ORIGINS	["https://leadpulse.vercel.app","http://localhost:3000"]

⚠️ Generate SECRET_KEY properly. Run this command in your terminal: openssl rand -hex 32 and paste the 64-character result.

4.3 Get your Railway URL

50. In the Railway dashboard, click your service → Settings → Networking.
51. Click Generate Domain. Railway assigns a URL like: https://leadpulse-production.up.railway.app
52. Copy this URL — you need it for the Meta webhook (Step 3.5) and the ALLOWED_ORIGINS setting above.

4.4 Verify the backend is running

```
# Replace with your actual Railway URL
$ curl https://leadpulse-production.up.railway.app/health

# Expected response:
{"status": "healthy"}
```

✅ If you see {"status": "healthy"}, your backend is live. Now go back and complete Step 3.5 (Meta Webhook setup).

Step 5 — Deploy the Frontend (Vercel)

Vercel deploys React/Vite apps in under 2 minutes directly from GitHub. Zero config required.

5.1 Create a Vercel project

53. Go to vercel.com and sign in with GitHub.
54. Click Add New → Project.
55. Find your leadpulse repository and click Import.
56. Set the Root Directory to: frontend
57. Vercel auto-detects Vite. Leave Framework Preset as Vite.

5.2 Set environment variables in Vercel

58. Before clicking Deploy, click Environment Variables.
59. Add the following variables:

Variable Name	Value
VITE_API_URL	https://leadpulse-production.up.railway.app (your Railway URL from Step 4.3)
VITE_FACEBOOK_APP_ID	your Facebook App ID (from Step 3.3)

60. Click Deploy. Vercel builds and deploys in ~60 seconds.
61. Copy your Vercel URL (e.g. https://leadpulse.vercel.app). You need it for two things:

5.3 Update Facebook Login redirect URI

Now that you have your Vercel URL, go back to Meta:

62. Open developers.facebook.com → Your App → Facebook Login → Settings.
63. Add to Valid OAuth Redirect URIs: https://leadpulse.vercel.app/settings
64. Click Save Changes.

5.4 Update Railway ALLOWED_ORIGINS

65. Go to Railway → your service → Variables.
66. Update ALLOWED_ORIGINS to include your Vercel URL:

```
["https://leadpulse.vercel.app","http://localhost:3000"]
```

67. Railway auto-redeploys when you save the variable.

✅ Your frontend is live! Visit your Vercel URL and you should see the LeadPulse splash screen.

Step 6 — Connect & Test Everything

With all three services deployed, run through this checklist to confirm the full flow works end-to-end.

6.1 Register and log in

68. Open your Vercel URL (e.g. https://leadpulse.vercel.app).
69. Click Get Started Free.
70. Fill in your name, business name, email, and password. Click Create Account.
71. You should be redirected to the Dashboard automatically. ✅

6.2 Connect your Facebook account

72. Click Settings in the sidebar.
73. Click Connect Facebook Account. You are redirected to Facebook.
74. Log in with the Facebook account that manages your ad pages.
75. Approve the permissions (leads_retrieval, pages_manage_metadata, pages_show_list).
76. You are redirected back to LeadPulse Settings. You should see your Page listed as LIVE. ✅

6.3 Test the webhook (real-time lead flow)

77. Go to developers.facebook.com → Your App → Webhooks.
78. Next to the leadgen subscription, click Test.
79. Meta sends a sample payload to your backend.
80. Switch to the LeadPulse Dashboard — you should see a new lead appear instantly with a green flash. ✅

6.4 Test the Simulate Lead button

81. On the Dashboard, click the ⚡ Simulate Lead button.
82. A mock lead appears at the top of the list with a flash animation. ✅
83. Click the lead to open the detail panel.
84. Change the status from New to Contacted. ✅
85. Add a note and click Save Note. ✅
86. Click WhatsApp — it should open wa.me with the lead phone number. ✅

6.5 Run the backend test suite

```
# On your local machine inside the backend folder:
$ cd backend
$ python -m venv venv && source venv/bin/activate
$ pip install -r requirements.txt
$ pytest tests/ -v

# Expected: all tests pass
  tests/test_auth.py         PASSED (8 tests)
  tests/test_webhook.py      PASSED (7 tests)
  tests/test_lead_service.py PASSED (7 tests)
```

Step 7 — Custom Domain (Optional)

Replace the Vercel and Railway subdomains with your own domain name (e.g. app.yourcrm.com).

7.1 Add custom domain to Vercel (Frontend)

87. In Vercel → your project → Settings → Domains.
88. Enter your domain, e.g. app.leadpulse.io, and click Add.
89. Vercel shows you a CNAME record to add to your DNS. Example:

```
Type:  CNAME
Name:  app
Value: cname.vercel-dns.com
```

90. Add this record in your domain registrar (GoDaddy, Namecheap, Cloudflare, etc.).
91. Wait 5–30 minutes for DNS to propagate. Vercel auto-provisions an SSL certificate.

7.2 Add custom domain to Railway (Backend)

92. In Railway → your service → Settings → Networking → Custom Domain.
93. Enter your API domain, e.g. api.leadpulse.io.
94. Railway shows you a CNAME record. Add it to your DNS:

```
Type:  CNAME
Name:  api
Value: XXXX.railway.app
```

7.3 Update all references to the new domains

- Update VITE_API_URL in Vercel to https://api.leadpulse.io
- Update ALLOWED_ORIGINS in Railway to ["https://app.leadpulse.io"]
- Update the Meta webhook Callback URL to https://api.leadpulse.io/webhook/facebook
- Update the Facebook Login redirect URI to https://app.leadpulse.io/settings

Step 8 — Production Security Checklist

Before sharing your app publicly, go through every item in this checklist.

Backend

- SECRET_KEY is a random 64-character hex string generated with openssl rand -hex 32
- DEBUG=false and ENVIRONMENT=production are set in Railway variables
- API docs (/docs) are hidden — they only appear when DEBUG=true
- ALLOWED_ORIGINS contains only your actual frontend domains, not wildcards
- SUPABASE_SERVICE_ROLE_KEY is set only in Railway — never in the frontend or git
- FACEBOOK_APP_SECRET is set only in Railway — never exposed to the browser

Supabase

- Row Level Security (RLS) is enabled on all tables (the schema does this automatically)
- The database password you set during project creation is stored somewhere safe
- You are NOT using the service_role key on the client side anywhere

Meta App

- The webhook Verify Token matches FACEBOOK_WEBHOOK_VERIFY_TOKEN in Railway exactly
- The webhook is subscribed to the leadgen field
- Valid OAuth Redirect URIs includes your production frontend URL
- Your Meta app is in Development mode initially — submit for App Review to access real leads in production. While in Development, only app admins/testers can connect Facebook pages.

Frontend

- VITE_API_URL points to your production Railway URL, not localhost
- VITE_FACEBOOK_APP_ID is your production app ID

⚠️ Meta App Review: To collect real leads from real Facebook users, your Meta app must pass App Review. Go to App Review → Requests in the Meta App Dashboard and request leads_retrieval permission. This can take 1–5 business days.

Step 9 — Pushing Updates

After the initial deployment, pushing code updates is simple. Both Railway and Vercel auto-deploy on every git push to main.

9.1 Deploy a code change

```
# Make your changes, then:
$ git add .
$ git commit -m "Your change description"
$ git push origin main

# Railway and Vercel both detect the push and auto-redeploy.
# Backend deploy: ~2 minutes
# Frontend deploy: ~60 seconds
```

9.2 Use the Makefile shortcuts

Command	What it does
make dev	Start full local stack with Docker (API + Redis + Vite + ngrok)
make backend	Run FastAPI dev server locally without Docker
make frontend	Run Vite dev server locally without Docker
make test	Run all 22 backend tests
make build	Build frontend for production (output: frontend/dist/)
make lint	Lint Python code with ruff
make clean	Remove __pycache__, .pyc files, and Vite build cache

9.3 View backend logs

95. Go to Railway → your service → Logs tab.
96. All requests are logged with method, path, status code, and duration.
97. Webhook events log as: 📥 Webhook received: 1 lead(s)
98. New leads log as: ✅ Lead saved: {uuid} — {name}

Troubleshooting

Webhook verification fails

Meta calls your GET /webhook/facebook endpoint during setup. If it fails:

- Confirm your Railway backend is running: curl https://your-railway-url.railway.app/health
- Confirm FACEBOOK_WEBHOOK_VERIFY_TOKEN in Railway exactly matches the Verify Token you entered in Meta
- Check Railway logs for the verification request

Leads are not appearing in real time

- Check that the leadgen field is subscribed in Meta Webhooks dashboard
- Check Railway logs for webhook POST events arriving
- Confirm the Facebook Page is connected in LeadPulse Settings (must show LIVE)
- The SSE stream (/api/v1/leads/stream) must be open in the browser — check the Network tab in DevTools

Facebook Connect fails (OAuth error)

- Confirm the redirect URI in Facebook Login → Settings exactly matches: https://your-vercel-url.vercel.app/settings
- Confirm VITE_FACEBOOK_APP_ID in Vercel matches the App ID in Meta
- Ensure your Meta app is not in maintenance mode

CORS errors in browser console

- Update ALLOWED_ORIGINS in Railway to include your exact frontend domain (no trailing slash)
- After updating, Railway will auto-redeploy — wait 1–2 minutes

Database errors (500 from API)

- Confirm you ran the full supabase_schema.sql in the Supabase SQL Editor
- Check SUPABASE_URL and SUPABASE_SERVICE_ROLE_KEY in Railway are correct — copy them again directly from Supabase Settings → API
- In Supabase, go to Table Editor and confirm the leads, users, workspaces, notes, and connected_pages tables exist

Need help?

If you get stuck on any step:

- Check the Railway logs for the exact error message
- Check the browser DevTools Console and Network tabs for failed requests
- The API interactive docs (available at your Railway URL/docs when DEBUG=true) show all endpoints and let you test them directly

> ✅ Most issues are caused by a mismatch between environment variables across services. Double-check that your URLs and keys are copy-pasted exactly, with no trailing spaces or slashes.